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Next-Latent Prediction Transformers Learn Compact World Models

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Abstract

Transformers replace recurrence with a memory that grows with sequence length and self-attention that enables ad-hoc lookups over past tokens. Consequently, they lack an inherent incentive to compress history into compact latent states with consistent transition rules. This often leads to learning solutions that generalize poorly. We introduce **Next-Latent Prediction (NextLat)**, which extends standard next-token training with *self-supervised* predictions in the latent space. Specifically, NextLat trains a transformer to learn latent representations that are predictive of the next latent state given the next token. Theoretically, we show that these latents provably converge towards *belief states*, compressed information about the history necessary to predict the future. This simple auxiliary objective injects a recurrent inductive bias into transformers while leaving their architecture, parallel training efficiency, and inference unchanged. NextLat effectively encourages transformers to form compact internal world models with coherent belief states and transition dynamics—crucial properties not guaranteed by standard next-token prediction alone. Empirically, across benchmarks in world modeling, reasoning, planning, and language modeling, NextLat demonstrates significant gains over standard next-token prediction and other baselines in downstream accuracy, representation compression, and lookahead planning. Furthermore, NextLat enables *variable-length self-speculative decoding*, accelerating inference by up to $3.3\times$ in language modeling. NextLat offers a simple yet effective paradigm for learning compact, predictive representations in transformers that generalize better. Our code is available at https://github.com/JaydenTeoh/NextLat.

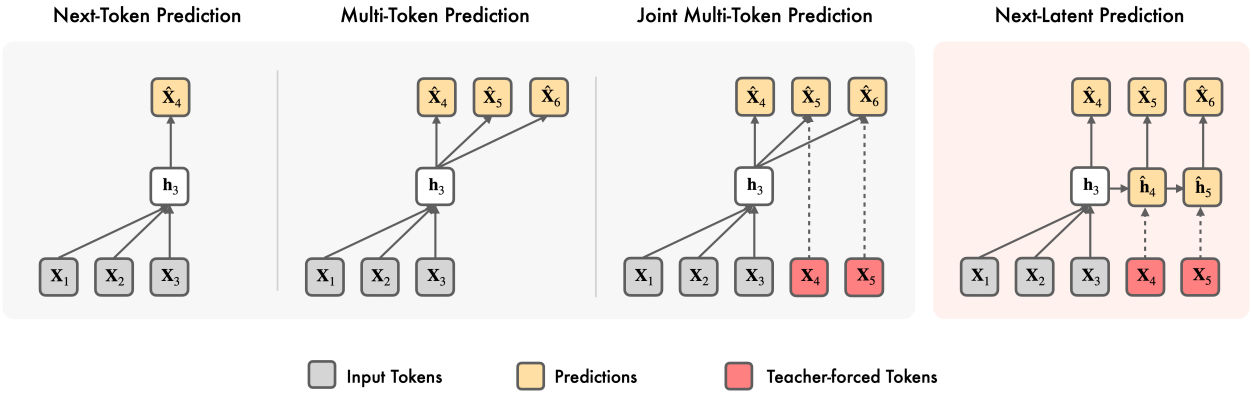


Figure 1: **Illustration of different predictive mechanisms.** Other methods supervise only in token space, leaving intermediate latent representations implicit. In contrast, NextLat explicitly trains the model to predict the next latent state (h_{t+1}) given the current latent state (h_t) and next token (X_{t+1}). Token supervision is then applied to the latent prediction ($\hat{h}_{t+1}$). Therefore, accurate multi-token predictions emerge from faithful latent dynamics modeling.

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1 Introduction

Ptolemy's geocentric model was able to accurately predict observations of the solar system from Earth's viewpoint, yet it was structurally convoluted: at times implying that the Moon came twice as close to Earth as other times. It was later supplanted by Copernicus's simpler, more compact heliocentric model, which provided accurate predictions that generalized beyond Earth's perspective. In learning theory, it is well known that simpler explanations of training observations tend to generalize better [Blumer et al., 1986, Langford and Schapire, 2005].

Modern transformers [Vaswani et al., 2017] stand in contrast to this principle. By replacing recurrence with a memory that scales with sequence length and self-attention that enables flexible lookups over past tokens, they achieve exceptional parallelization and predictive power. Yet, this very capability removes any inherent pressure to compress history into compact latent representations with consistent update rules. As a result, transformers often learn complex, task-specific shortcuts that fit the training data well but generalize poorly [Anil et al., 2022, Dziri et al., 2023, Liu et al., 2023, Wu et al., 2024]. How can we encourage transformers to form simpler, more principled explanations that avoid such shortcuts? A natural approach is to reinstate a key property of recurrent models: the ability to learn *compact world models* that channel future prediction through compressed representations of the past. We will show that this inductive bias can be reintroduced while also retaining the parallel training efficiency of transformers.

In this paper, we introduce **Next-Latent Prediction (NextLat)**, which extends the standard next-token prediction objective with self-supervised predictions in latent space. NextLat jointly trains a transformer and a latent dynamics model: the transformer learns to encode past tokens into compact latent summaries such that the dynamics model can predict the transformer's next latent state given only the current latent state and the next token (i.e., the "action"). This objective encourages the transformer to form a compact internal world model with coherent recurrent-like dynamics, while avoiding the sequential processing overhead of recurrent architectures. Importantly, NextLat leaves the transformer's architecture, inference procedure, and parallel training efficiency unchanged, adding only a lightweight auxiliary loss on latent representations during training. By augmenting sparse one-hot token targets with dense latent-state supervision, NextLat provides richer learning signals and improves data efficiency compared to next-token prediction and other token-level supervision methods. Our approach is inspired by the *self-predictive learning* paradigm in reinforcement learning (RL), a family of algorithms that learn representations by minimizing the prediction error of their own future latent states [Tang et al., 2023, Ni et al., 2024].

Beyond representation learning, latent dynamics also provides an inference advantage: future states can be recursively predicted directly in latent space without invoking the main transformer. This recursive multi-step lookahead using the lightweight dynamics model enables *variable-length self-speculative decoding*: the model can speculate a flexible number of future tokens and accelerate inference without requiring separate multi-token prediction heads.

The core contributions of this paper are threefold. First, we establish a theoretical foundation showing that NextLat provably shapes transformer representations into *belief states*—compact summaries of past information sufficient for predicting future observations. Such representations are important for planning and generalization, yet are not guaranteed to emerge from next-token prediction alone. Second, we present a practical implementation of NextLat that preserves the transformer's architecture, inference procedure, and parallel training efficiency. Finally, we empirically demonstrate NextLat's effectiveness across diverse domains spanning world modeling, reasoning, planning, and language modeling. Our results show that NextLat improves representation compactness, lookahead planning, and downstream accuracy over standard next-token prediction and other baselines. In language modeling, NextLat achieves up to $3.3\times$ faster inference through variable-length self-speculative decoding. Together, these results position NextLat as an efficient framework for learning compact, predictive, and generalizable representations in transformers.

2 Related Work

We are motivated by a long line of prior works in representation learning for prediction and control. Of close interest are self-supervised learning methods, belief states for decision making, and world models.

Self-Supervised Learning. Self-supervised learning (SSL) is a framework for learning from unlabeled data, where a model generates its own supervisory signals from the structure of raw inputs. Across modalities such as vision, audio, and time series, SSL has proven highly effective for pretraining useful features, enabling downstream transfer that rivals, or even surpasses, models trained on labeled data [Liu et al., 2022, Balestriero et al., 2023, Zhang et al., 2024]. There are several approaches to SSL. Our method falls under *self-predictive representation learning*, which jointly learns latent representations and a transition function that models how these representations evolve over a sequence. Self-prediction has driven state-of-the-art advances in RL [Gelada et al., 2019, Zhang et al., 2020, Ye et al., 2021, Schwarzer et al., 2021, Hansen et al., 2022]. However, latent-space SSL remains underexplored in language modeling. A recent effort, LLM-JEPA [Huang et al., 2025], minimizes distances between embeddings of paired text–code data, but relies on manually curated pairs and therefore does not generalize to raw text. In contrast, our method introduces a fully self-supervised latent prediction objective requiring no paired data, making it broadly applicable for training transformers across arbitrary sequence modeling domains and data sources.

Belief States. In both sequence modeling and RL, models must reason over long histories of observations. To mitigate this curse of dimensionality, prior work focuses on compressing history into latent representations that capture all information necessary for future prediction. In RL, this latent summary is formalized by Kaelbling et al. [1998] as a *belief state*, defined as: *"a sufficient statistic for the past history ... no additional data about its past actions or observations would supply any further information about the current state of the world"*. In stochastic control, the same notion of sufficient statistics appears as "information state" [Striebel, 1965]. The idea of sufficient statistics is also key to learning state abstractions in RL [Li et al., 2006]. While recurrent neural networks naturally enforce such compression, transformers have no such constraint—their internal state, or memory, grows linearly with sequence length. Recently, Belief State Transformers (BST; [Hu et al., 2025]) extended the notion of belief states to transformers, and demonstrated benefits in planning tasks. Compared to BST, NextLat learns belief states without requiring a separate transformer and is much more computationally efficient. We compare these methods further in Section 5.

World Models. Loosely, a world model is an internal predictive model of how the world works, with varying interpretations across cognitive science [Craik, 1967, Johnson-Laird, 1983], neuroscience [Miall and Wolpert, 1996, Friston, 2010], control theory [Francis and Wonham, 1976, Conant and Ross Ashby, 1970] and reinforcement learning [Sutton, 1991, Schmidhuber, 1990, Ha and Schmidhuber, 2018]. Whether transformer language models implicitly learn world models remains a debate; some studies report emergent world understanding [Patel and Pavlick, 2022, Li et al., 2023, Gurnee and Tegmark, 2024], while others find incoherent world structure [Vafa et al., 2024, 2025]. However, successful world modeling approaches such as *MuZero* [Schrittwieser et al., 2020] for achieving superhuman performance in video and board games, *Dreamer* [Hafner et al., 2019, 2021, 2025] for model-based RL, and *Genie* [Bruce et al., 2024] for interactive video generation share a common principle: they learn a latent dynamics model that takes a latent state (an encoding of past observations), an action, and predicts the next latent state. Yet, learning such latent dynamics for transformer-based language modeling remains underexplored. NextLat addresses this gap by explicitly learning a latent dynamics model that governs how the transformer's latent states evolve given new tokens (i.e., "actions"), enabling the transformer to learn compact latent abstractions of the world with consistent dynamics.

Beyond Next-Token Prediction. In the domain of language modeling, a growing body of work has highlighted the myopic nature of the next-token prediction objective, which limits the model's capability in downstream tasks such as planning and reasoning [Bachmann and Nagarajan, 2024, Nagarajan et al., 2025]. Recent works have also found improvements from richer supervision signals that predict further into the future [Gloeckle et al., 2024, Liu et al., 2024, Hu et al., 2025, Ahn et al., 2025]. However, these approaches operate predominantly in the token space. NextLat takes a different approach: it shifts prediction into the latent space, enforcing coherent dynamics over the model's latent representations rather than its token outputs. As we discuss later in Section 3.1, this latent-space supervision provides richer gradient signals than token-level supervision.

Speculative Decoding. Speculative decoding [Leviathan et al., 2022, Chen et al., 2023] accelerates inference by using a lightweight draft model to propose multiple tokens, which are then verified in parallel by a target model. Closely related, Li et al. [2024a,b] train a draft model on top of a frozen language model to predict its high-level latent features.

Unlike our approach, this is done post-hoc, whereas we study latent prediction as a pretraining objective to directly shape representations. Moreover, their draft model relies on full attention over past features and therefore does not learn belief states, i.e., compact summaries of past tokens. *Self-speculative decoding* removes the need for a separate draft model by having a single model act as both draft and verifier. A common implementation uses the multi-token prediction (MTP) objective, where the model is trained to predict multiple future tokens that can be verified in parallel [Gloeckle et al., 2024]. However, MTP operates in token space and is therefore *typically* constrained to a fixed speculative horizon determined during training. In contrast, NextLat learns a latent dynamics model that can be recursively composed in latent space, enabling *variable-length self-speculative decoding* and faster inference.

3 Methodology: Next-Latent Prediction

In this section, we introduce a simple, yet powerful, method for learning belief states in transformers via next-latent prediction (or more specifically, via *next-hidden state prediction*[1]). We begin by defining belief states in sequence modeling.

Definition 3.1 (Belief states in sequence modeling)**.** Let $X_{1:T}$ denote a token sequence $X_1, \ldots, X_T$. A random variable $\mathbf{b}_t = g(X_{1:t})$ is a *belief state* for the history $X_{1:t}$ if, for every bounded measurable function f of the future,

$$\mathbb{E}[f(X_{t+1:T}) \mid \mathbf{b}_t] = \mathbb{E}[f(X_{t+1:T}) \mid X_{1:t}] \quad \text{a.s.}$$

Equivalently, $\mathbf{b}_t$ is a *sufficient statistic* [Striebel, 1965] of the history $X_{1:t}$ for predicting the future tokens, i.e., from which we can sample from the distribution $\mathbb{P}(X_{t+1:T} \mid X_{1:t})$. Next, we describe how next-latent prediction enables transformers to learn belief states and improves data efficiency.

3.1 Why Next-Latent Prediction?

Here we analyze an idealized next-latent prediction transformer which successfully optimizes both next-token prediction and next-latent prediction with respect to an underlying data distribution.

Theorem 3.2. *Consider the joint learning of three components:*

1. *a transformer with parameters θ that produces hidden states $\mathbf{h}_t$ at each time step t,*
2. *an output head p_θ modeling the next-token distribution, and*
3. *a latent dynamics model p_ψ modeling the transition dynamics of the transformer's hidden states.*

If NextLat successfully optimizes the following objectives:

$$\textbf{\textit{(Next-Token Consistency):}} \quad p_\theta(X_{t+1} \mid \mathbf{h}_t) = \mathbb{P}(X_{t+1} \mid X_{1:t}), \tag{1}$$

$$\textbf{\textit{(Transition Consistency):}} \quad p_\psi(\mathbf{h}_{t+1} \mid \mathbf{h}_t, X_{t+1}) = \mathbb{P}(\mathbf{h}_{t+1} \mid X_{1:t+1}), \tag{2}$$

then $\mathbf{h}_t$ must be a belief state for the sequence $X_{1:t}$. Note that the right-hand side of Equation 2 is the transition law induced by the transformer's weights[2]*.*

Proof Sketch. A formal proof by backward induction is provided in Section B. Intuitively, optimizing for next-token (Equation 1) and transition (Equation 2) consistency ensures existence of measurable maps, i.e., p_θ and p_ψ, that allow recursive decoding of future tokens from $\mathbf{h}_t$:

$$\mathbf{h}_t \xrightarrow[\text{decode token}]{p_\theta} X_{t+1} \xrightarrow[\text{update state}]{p_\psi} \mathbf{h}_{t+1} \xrightarrow[\text{decode token}]{p_\theta} X_{t+2} \xrightarrow[\text{update state}]{p_\psi} \mathbf{h}_{t+2} \cdots \xrightarrow{p_\theta} X_T.$$

[1] In the sequence modeling literature, intermediate latent representations are often referred to as "hidden states". To disambiguate, we use the term "latent state" to broadly refer to learned representations within the transformer's residual stream, and "hidden state" to refer to a subset of this representation—specifically, the final layer's output at each time step (i.e., the pre-logit activations).

[2] We adopt a probabilistic formulation to retain generality with respect to stochastic transformer models, e.g. Fleuret [2025].